



Executive Summary Template

Answer Recommended Solution

- Implement a targeted 20% discount strategy for SME customers with a predicted churn probability >50%. This surgical approach maximizes retention while preserving margin from the loyal 90% of the customer base.

Insert Situation

- PowerCo, a major energy utility, is facing increasing customer attrition within the SME segment amidst a highly competitive and evolving energy market. Powerco has a problem with customer churn; they believe it is caused by customers' price sensitivities. One possible solution is to provide 20% off to customers who are most likely to start leaving.

Insert Complication

- High churn rates threaten long-term profitability. Initial hypotheses suggested price sensitivity as the primary driver, but analysis showed low correlation between price changes and churn behavior.

Insert Question

- Can a predictive model accurately identify at-risk customers to enable a cost-effective retention strategy? Result: A Random Forest Classifier achieved 90% accuracy, identifying tenure and consumption as key churn drivers.



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Machine Learning Modeling:

- After Data cleaning, EDA and Feature engineering, I applied Random Forest Classifier. Random Forest Classifier model has been built to predict customers' churn probability, achieve in accuracy of 0.90 and Precision score of 0.91 on test set.

Insights:

- Nearly 10% (9.7%) of the customers have churned and 90% of the customers have not churned.
- Net margin on power subscription and consumption over 12 months is a top driver for churn.
- Forecasted bill of meter rental for the next 2 months also is an influential driver.
- Time seems to be an influential factor, especially the number of s they have been active, their tenure and the number of months since they updated their contract.